

SautiLedger: Code-Switch Speech Benchmark

Sahara CodeSwitch Africa Challenge, Phase 2 · David Agbolade · 7 speech systems on code-switched African market and broadcast speech, scored on whether **the money** survives, not only the words · Full report, data card and code: github.com/dagbolade/sautiledger (bench/results/REPORT.md, bench/DATASET.md)

Summary.

- **No model wins everywhere, and in our sample the leader flips with the language** (a direction, not yet a proven ranking: see section 8). On Nigerian Pidgin/Yoruba market speech, MAI-Transcribe-2 records the most transactions exactly (60%), ahead of GPT-4o-transcribe (53%) and Sahara v2.5 (47%), all three with 0% corrupted amounts. On Shona the order inverts: Sahara v2.5 and Meta's open omnilingual-ASR tie at 27% exact with 0% corrupted, double the best frontier system, while three frontier systems produce wrong parser amounts.
- **WER hid the Shona failure; the task metric exposed it.** Sahara's Shona WER (0.566) is slightly *better* than its Pidgin WER (0.574), yet its transaction accuracy falls from 47% to 27%. It keeps the Shona words and loses the English price ("*ne five dollars fifty*" → ne50).
- **Sahara v2.5 has the lowest WER on all three frozen tiers**, but not everywhere: Chirp-3 is better on AfriSwitch Yoruba and Hausa, and on a supplementary Yoruba-English farm tier from a second speaker Sahara is mid-field (section 7).
- **Safety has to be engineered, not inferred from accuracy.** A plausible-but-wrong transcript is more dangerous than a garbled one, because only the plausible one parses. Our grammar turns uncertainty into a question; on Shona, Sahara produced 15/15 amount-safe outcomes.

1. Data

Tier	Source	Languages	Clips	Audio	Speakers	Ground truth
tier-a market	Recorded for this benchmark; sentences corrected by a native speaker before recording	Nigerian Pidgin + Yoruba + English	15	1.1 min	1 Nigerian male, own phone	Transcript + ledger entry (intent, item, qty, unit, amount)
tier-sh market	Recorded 7 Sep by a native Shona speaker, who also rewrote the sentences	Shona + English, prices in USD	15	1.2 min	1 Zimbabwean female, own phone	Transcript + ledger entry (amounts in cents)
tier-b broadcast	intronhealth/AfriSwitch test split, first rows per language (CC BY-NC-SA 4.0), release before 7 Sep	Pidgin 16 · Swahili 12 · Yoruba 6 · Hausa 6 (each + English)	40	8.2 min	many	Transcript only

Total: 70 clips, 10.5 audio-minutes. The 30 native clips cover 15 transactions, 5 cases where the correct response is to *ask* (e.g. a sale with no price), 4 ledger queries, 4 corrections and 2 daily summaries. **Preprocessing:** phone audio converted to 16 kHz mono 16-bit PCM WAV, with quality gates (1-15 s, not silent); AfriSwitch audio used as published. Text normalised for scoring: lower-case, punctuation removed, whitespace collapsed, diacritics folded (so a model is not penalised for correct Yoruba tone marks the reference lacks). Raw, unnormalised scores are in the full report. **Frozen:** the manifest sha256 50e5e064...d4377aeb was fixed before the first model ran, and no clip was added or dropped afterwards.

Field use (evidence, not a scored tier). The live app was used on phones from 26 Aug to 12 Sep: 39 device sessions, 217 utterances (91 spoken), 50 ledger entries and 91 clarifying questions. These counts include our own test sessions, which cannot be cleanly separated from testers'. Nine conversation scenarios (section 6) replay real sessions word for word: 6 now complete after the fixes they prompted, and 3 still fail, none with a wrong amount.

2. Systems and metrics

Systems (September 2026): Intron **Sahara v2.5** (sync API, explicit use_language_asr_input); **MAI-Transcribe-2** (Microsoft), **GPT-4o-transcribe** (OpenAI), **Chirp-3** (Google) and **Parakeet-TDT 0.6B v3** (NVIDIA), all via OpenRouter; **Whisper-large-v3** (local) and Meta **omnilingual-ASR CTC-300M** (local, open weights). Every transcript is cached, so all scores can be recomputed without new API calls.

Metric	What it measures	Why it is appropriate here
Transaction exact	Transcript fed through the app's real parser; every labelled field of the parser result correct; this does not execute a database write	The task is bookkeeping. At Intron's 28 Aug masterclass, the metric recommended for a financial app was "how many transactions were actually correctly recorded".
Amount corrupted	The parser produced a <i>wrong</i> amount; commit gates and confirmation are measured separately	The one error a ledger cannot tolerate. It is kept separate from amount safe (right or the app asked), because asking is safe.
Numeric accuracy	Every number in the reference survives transcription	Isolates the digits a ledger depends on, independent of our parser.
WER / CER	Word and character error rate	Comparable with public leaderboards. CER is fairer for morphologically rich Bantu languages, where one wrong affix costs a whole word.

3. Downstream task results (native market speech)

Model	tier-a · Pidgin / Yoruba / English (n=15)			tier-sh · Shona / English (n=15)		
	Transaction exact	Amount corrupted	Numeric acc.	Transaction exact	Amount corrupted	Numeric acc.
Sahara v2.5	47%	0%	67%	27%	0%	40%
MAI-Transcribe-2	60%	0%	80%	13%	13%	53%
GPT-4o-transcribe	53%	0%	80%	13%	7%	67%
Parakeet-TDT	33%	0%	80%	7%	13%	93%
Chirp-3 (14/15 on tier-a)	29%	7%	71%	13%	0%	53%
Whisper-large-v3	27%	0%	67%	7%	13%	60%
omnilingual CTC-300M	13%	0%	53%	27%	0%	47%

With n=15, one clip moves a percentage by about 7 points, so differences of one or two clips are not meaningful. A high numeric accuracy with a low transaction score (Parakeet on Shona, 93% vs 7%) means the digits were heard but the item or structure was lost.

4. WER / CER per language

Model	Pidgin market (15)		Shona market (15)		AfriSwitch Pidgin (16)		AfriSwitch Swahili (12)		AfriSwitch Yoruba (6)		AfriSwitch Hausa (6)	
	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER
Sahara v2.5	.574	.490	.566	.328	.232	.174	.295	.154	.896	.559	.356	.182
MAI-Transcribe-2	.663	.496	.869	.399	.294	.149	.370	.116	.862	.376	.790	.290
GPT-4o-transcribe	.652	.404	1.018	.402	.497	.423	.463	.325	.747	.556	.931	.699
Parakeet-TDT	.651	.413	1.131	.478	.370	.223	.765	.346	.909	.700	.855	.578
Chirp-3	.778*	.598*	.900	.467	.316	.204	.335*	.157*	.683	.599	.305	.140
Whisper-large-v3	.963	.710	1.232	.537	.335	.217	1.189	.728	.912	.637	.914	.700
omnilingual CTC-300M	.861	.542	.790	.342	.952	.887	.678	.383	.806	.410	.878	.552

Normalised mean per clip; lower is better; best per column shaded. * one clip returned no transcript and is excluded (14/15, 11/12). The Yoruba and Hausa columns have 6 clips each and are indicative only. Sahara transcribes spoken "five thousand five" as "5,500", which is exact for a ledger but counts as a word error, one reason WER understates usefulness here.

5. Strengths and weaknesses, per model per language

Model	Pidgin / Yoruba market	Shona market	Broadcast (AfriSwitch)
Sahara v2.5	+ Keeps Pidgin's perfective " <i>I don sell</i> " (a sale); 0% corrupted. – 3rd on transactions (47%); one clip lost a unit and a factor of ten (" <i>3 derica of rice 5,500</i> " → 3 of rice 500).	+ Best WER and CER; Shona words intact (<i>ndatengesa</i> , <i>matomatsi</i> , <i>chikwereti</i>); tied first, 0% corrupted. – The English price collapses into the numeral (ne50, etridos); lowest numeric accuracy (40%).	+ Best WER on Pidgin and Swahili. – Weak on Yoruba (.896); behind Chirp-3 on Hausa. No model-version field, so drift between our Aug and Sep runs cannot be attributed.
MAI-Transcribe-2	+ Best transactions (60%) with 0% corrupted. – Inverts the perfective into " <i>I don't sell</i> ", turning a sale into its denial.	– Collapses: 13% exact and 13% corrupted.	+ Best CER on Pidgin, Swahili and Yoruba. – High WER on Hausa (.790) and Yoruba.
GPT-4o-transcribe	+ 53% exact, 0% corrupted; often verbatim (" <i>five thousand five</i> "). – Once turned the perfective into " <i>I no sell</i> ".	– WER above 1; produced a wrong parser amount on per-unit pricing (" <i>two fifty hwani</i> ", 7%).	– Weakest cloud system on Hausa (.931); .497 on Pidgin, behind all but omnilingual.
Parakeet-TDT	+ 0% corrupted; very cheap (\$0.0015/min). – 33% exact; inverts the perfective.	– 7% exact and 13% corrupted despite the highest numeric accuracy (93%): digits heard, sentence lost.	+ Reasonable on Pidgin (.370). – Poor on Swahili, Yoruba and Hausa.
Chirp-3	– 29% exact, 7% corrupted; mangles market units (<i>derica</i> → L); inverts the perfective; one empty result.	+ 0% corrupted; keeps the amount (\$5.50). – Degrades the Shona verb and concord, so only 13% exact.	+ Best WER on Yoruba and Hausa; third on Pidgin (.316). – One empty Swahili result.
Whisper-large-v3	+ Offline, free. – Anglicises the speech and inverts the perfective; 27% exact.	– Hallucinated " <i>Thank you for watching</i> " on two clips; turned a spoken correction into the very amount being corrected away from (550); 13% corrupted.	+ Competitive on Pidgin (.335). – WER above 1 on Swahili.
omnilingual CTC-300M	+ Open weights; correct Yoruba tone marks. – 13% exact; hears the words but drops the digits.	+ Tied first (27%), 0% corrupted, second-best CER. – Phonetic spelling.	– Worst on Pidgin (.952). About 20-25 s per clip on a CPU laptop (1-2 min under memory pressure); not free without hardware.

6. Beyond transcription: the full loop

Conversation benchmark (agent + real ledger)

37 scripted sessions replayed through the real agent and SQLite ledger: clarifying questions, confirmations, rejections and repairs included. **16 of 19 tasks completed, none ending with a wrong amount. 18 of 18 controls** correctly did *not* complete. 17 of those are real, unscripted Shona market chatter (greetings, "*your tomatoes aren't fresh*", "*won't you give me a little extra?*"), replayed exactly as Sahara transcribed it: **0 ledger rows written**. Five scenarios start with a deliberately wrong amount and repair it. Wrong amounts are counted at every turn, not only at the end, so a repaired mistake still shows up.

TTS round-trip (Sahara TTS)

The app reads every entry back aloud. We synthesised its real confirmations and transcribed them with an independent ASR (Whisper-small). The first run caught a product bug: the app was **speaking its punctuation** ("*expense: fuel*" came back as "*expense call on*"), which corrupted the amount. After the fix: **hallucination 0.117 → 0.000, amount survival 91% → 100%** (Pidgin voice; Yoruba-accent voice 91% → 100%).

7. Qualitative findings

- **The meaning-reversal error.** Four of the seven systems transcribe Pidgin's perfective "*I don sell*" ("I have sold") as "*I don't sell*". WER counts that as one wrong word, but it reverses the transaction. Of the three such sentences in the corpus, Sahara v2.5 kept the meaning in all three; GPT-4o in two (the third became "*I no sell*", also a negation).
- **Which half of a code-switched sentence survives depends on the model.** On Shona, Sahara keeps the Shona and loses the English price; Chirp-3 does the opposite. A ledger can ask for a missing amount but cannot recover a missing item, an asymmetry WER cannot see.
- **The ranking does not transfer.** Benchmark only on Pidgin and you pick MAI-Transcribe-2, which corrupts 13% of Shona amounts; only on Shona and you pick Sahara and lose 13 points on Pidgin. Route by language, and measure per language and task.
- **A second speaker changes the picture again.** A supplementary tier (recorded 14 Sep, own hash, outside the frozen corpus): 15 Yoruba-English farm-trade lines from a second Nigerian speaker, labelled before recording. WER: Parakeet .660, GPT-4o .696, MAI .742, Sahara .747, Chirp-3 .763, omnilingual .807, Whisper .847, no gap significant; omnilingual's CER (.339) is significantly below Sahara's (.525, $p=.035$). Exact transactions are near zero for all: models fuse *mo ta* into *Mota* and anglicise items (*isu* → *issue*), yet amounts stay safe 80-93% of the time. The main surviving corruption is a pack gap: *ikan je* ("each") is unknown. References are her adapted script; two lines with a different spoken price are reported separately.
- **A registered prediction, partly falsified.** Before any Shona audio existed, we predicted (in a committed file) that Sahara would degrade on Shona, which is a supported language but not a documented code-switch pair. On WER we were wrong; on transactions we were right.

8. Limitations and product feedback

- **Small and confounded.** 15 clips per native tier, one speaker each, so language is confounded with voice, microphone and room. Our explanation for the Pidgin/Shona reversal (Pidgin is lexically close to English, Shona is not) is a **hypothesis**; testing it would need the same speakers across languages. The speech is read, not live trade.
- **Statistical confidence.** Paired exact McNemar/sign tests with bootstrap 95% intervals (python -m bench.significance). **No transaction-accuracy gap is significant at n=15** (MAI vs Sahara: 4 Pidgin clips differ, 3-1, $p=.63$). **What holds:** Sahara's Shona WER is below GPT-4o ($p=.04$), Parakeet ($p=.01$), Chirp-3 ($p=.02$) and Whisper ($p<.001$), and its broadcast WER below MAI's ($p=.04$).
- **Script, not verbatim.** In 4 of 15 Pidgin clips the speaker added openers ("*How far na*") the reference lacks, inflating tier-a WER for all models. Without them Sahara's WER falls from .574 to .358 and it stays first; only two adjacent pairs swap. Labels are unaffected; the frozen references are unchanged.
- **Moving targets and a null result.** Sahara's output on identical audio changed between 5 Aug and 9 Sep (broadcast WER .424 → .369) with no version field; AfriSwitch was republished on 7 Sep, after our sample. `use_disable_llm_corrections=True` changed nothing on all 70 clips. A **re-run on 14 Sep**, after Intron made the language parameter mandatory, returned byte-identical transcripts from Sahara, Chirp-3 and Parakeet; GPT-4o changed 60 of 70 on the same audio, moving its Pidgin exact rate from 53% to 47%.
- **Feedback to Intron:** add a model-version field; reject unknown `use_*` fields (a misspelt language parameter silently fell back to English); make Shona a code-switch pair (AfriSwitch already has 3.86 h at CMI 24.55); fix streaming COMMIT, which returns INPUT_ERROR.

9. What we did with the result

SautiLedger runs on **Sahara v2.5** for speech input and read-back, and the benchmark shows why that is a reasoned choice rather than a default. Sahara has the lowest WER on the three frozen tiers, ties for first on Shona with no corrupted amounts, and was the only system to transcribe all three Pidgin perfective sentences with the verb intact and not negated. It is also the only vendor here that provides TTS in the same Nigerian register, which the spoken read-back depends on. It is **not** the best transcriber of Pidgin transactions: MAI-Transcribe-2 is. Our engineering answer is to make the **ledger** safe whatever the ASR does: one gate sits in front of every write, no amount is recorded unless it appears in the transcript, and the app asks when unsure. It cannot undo an ASR error that turns one plausible amount into another, which is why every entry is read back. The next step the data points to is **per-language routing**, and a pack entry for Yoruba *ikan je*.

Reproduce: `python -m bench.run --score-only` rebuilds every score from cached transcripts; `python -m bench.conversations` replays the sessions. `python -m bench.supplementary --score-only` scores the farm tier. All 45 native clips (three speakers) are published with each speaker's consent, with transcripts and ledger labels, at huggingface.co/datasets/dagbolade/sautiledger-market-speech; AfriSwitch audio is not redistributed. Hosted-demo privacy: audio and read-back text go to Sahara, unparsed text may go to a Hugging Face model; both are listed in the app.